Reclaim

Technical Design Document · First-5-Days Recovery Assistant for Cyber-Fraud Victims

EVENT	DOMAIN	VERSION	STATUS
Founder's Code 2026	Cybersecurity & Digital Safety	1.0 · 13 Oct 2026	Prototype design

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1. Purpose and scope

This document describes the technical design of **Reclaim**: a mobile-first web application that guides a victim of UPI or banking fraud through the three official reports — the **1930** helpline, a complaint on the **National Cyber Crime Reporting Portal (NCRP)**, and a written report to their **bank** — and tracks the deadlines that decide whether money can be held and whether the victim qualifies for compensation. A companion **bank dashboard** gives the bank's fraud team a live view of incoming, complete cases.

It covers the hackathon prototype in detail and the intended production architecture at design level. Build sequencing is in the separate Implementation Plan.

2. Problem and requirements

2.1 Problem

- About **51%** of UPI-fraud victims file no official complaint; users describe reporting as complex (LocalCircles survey, 32,000+ responses, Jun 2025).
- Recovery is low when reporting is late — Odisha recovered about **2.7%** of ~₹370 crore lost in 2023–25, with police citing delayed reporting.
- Recovery needs three separate actions in three places; the 1930 call issues a 14-digit acknowledgement that must then be used to complete the NCRP complaint.
- From **1 Jan 2027**, RBI's Third Amendment Directions (2026) compensate small-value frauds (loss ≤ ₹50,000; 85% of net loss, max ₹25,000; once per lifetime) **only if** the fraud is reported to both the bank and NCRP/1930 within **5 calendar days**. Banks must resolve domestic fraud complaints within **45 days**.

2.2 Functional requirements

ID	Requirement
FR-1	Accept a payment screenshot or bank SMS and extract amount, UTR, date-time, bank, payment app, method and payee.

ID	Requirement
FR-2	Let the user confirm or edit every extracted field; support manual entry and multiple transactions.
FR-3	Classify the scam type from the user's description, with a confidence score.
FR-4	Evaluate RBI compensation eligibility and compute the 5-day and 45-day deadlines.
FR-5	Generate a 1930 call script, an NCRP complaint description (≥ 200 characters) and a bank dispute letter.
FR-6	Record acknowledgement numbers for each report and show a case timeline.
FR-7	Produce a downloadable case pack (PDF).
FR-8	Show the bank a live list of cases, deadline status and scam-type distribution.
FR-9	Offer English and Tamil interfaces, and voice input for the description.

2.3 Non-functional requirements

ID	Requirement	Target (prototype)
NFR-1	Demo reliability	Full flow works with no internet and no AI key (cached fixtures, manual path)
NFR-2	Extraction latency	Cache < 100 ms; AI call bounded by a 12 s timeout
NFR-3	Privacy	No OTP/PIN/password ever collected; account numbers masked; no images or descriptions written to logs
NFR-4	Usability	Usable at 360 px width; tap targets ≥ 44 px; one primary action per screen
NFR-5	Correctness	Eligibility rules covered by unit tests; all times stored in UTC, shown in IST

2.4 Explicit non-goals

Reclaim does not file complaints on anyone's behalf, does not access bank accounts, does not promise refunds, and never handles or charges victims' money.

3. System overview

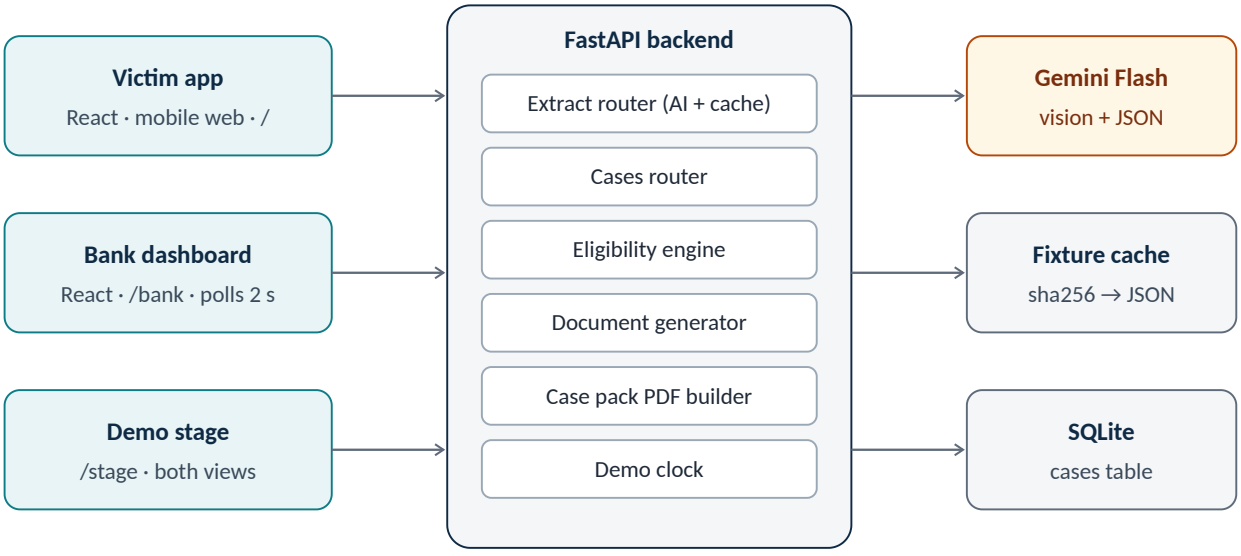


Figure 1. Prototype architecture — one backend process, one SQLite file, three web views.

Layer	Technology	Reason
Frontend	React, Vite, TypeScript, Tailwind CSS, React Router, Zustand, lucide-react	Fast to build, typed, mobile-first styling
Backend	Python, FastAPI, Pydantic, SQLModel, Uvicorn	Typed request/response models, auto docs at `/docs`
Storage	SQLite	Zero setup; single file; adequate for prototype load
AI	Gemini Flash (free tier) via `google-genai`	Reads images and returns schema-constrained JSON
PDF	ReportLab with a TTF font that includes the ₹ glyph	Deterministic server-side rendering
Voice	Browser Web Speech API (`en-IN` , `ta-IN`)	No key, no server cost

4. Component design

Component	Responsibility	Key interfaces
Extract router	Accepts image + description; runs the evidence pipeline (§9); returns fields, per-field confidence, scam type	`POST /api/extract`
Cases router	Creates, reads and updates cases; recomputes eligibility on every read; marks completion	`/api/cases` , `/api/cases/{id}`
Eligibility engine	Pure function implementing RBI rules (§8); no I/O; fully unit-tested	`evaluate(...) → Eligibility`
Document generator	Deterministic templates for the three official texts and follow-up letters (§10)	`GET /api/cases/{id}/docs`
PDF builder	Renders the case pack	`GET /api/cases/{id}/pack.pdf`
Demo clock	Provides `now()` ; in demo mode offsets real time from a fixed start (default 12 Jan 2027, 8:01 PM IST)	`clock.now()` , `POST /api/demo/clock`
Victim app	Screens S1–S9: safety check, upload, confirm, case home with countdown, three report steps, done, follow-up	REST over JSON
Bank dashboard	KPI tiles, live case table, deadline colouring, scam-type bars, case drawer	Polls `/api/cases` and `/api/stats`

5. Core flows

5.1 Victim journey (request sequence)

#	User action	Frontend	Backend
1	Opens link, answers safety question	S1 → S1b (if ongoing) → S2	—
2	Uploads screenshot, describes incident	Shows "Reading your screenshot…"	`POST /api/extract` → cache → AI → manual
3	Confirms or edits fields	S3 highlights fields with confidence < 0.7	`POST /api/cases` → stores case, returns eligibility
4	Sees plan and countdown	S4 ticks every second from `deadline_at`	`GET /api/cases/{id}` , `/docs`
5	Calls 1930, enters 14-digit ack	S5 validates `^\d{14}$`	`PATCH` sets `ack_1930` , `ack_1930_at`
6	Files on NCRP, enters ack	S6 copy box with character count	`PATCH` sets `ack_portal`
7	Emails bank, enters ticket no.	S7 `mailto:` pre-filled	`PATCH` sets `bank_ref` ; all three done → `completed_at` , status `complete`
8	Downloads case pack	S8	`GET /pack.pdf`

#	User action	Frontend	Backend
9	(Bank) sees the case	Dashboard row appears, highlights, turns green	`GET /api/cases` every 2 s

5.2 Case lifecycle

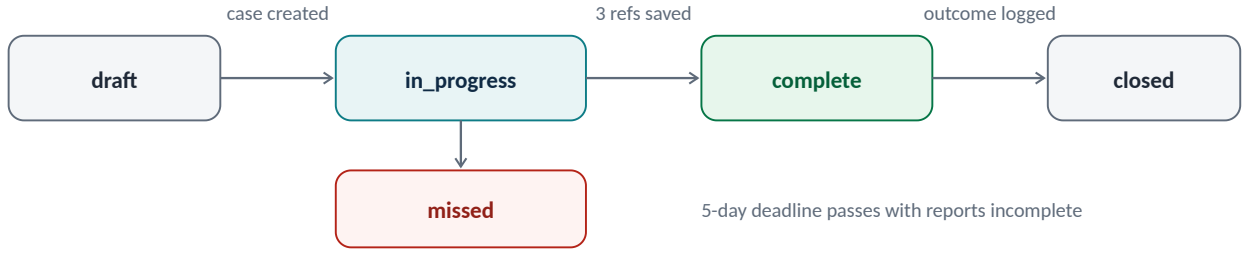


Figure 2. Case states. A `missed` case keeps all features; only the compensation message changes.

6. Data model

```
Case
  id                 TEXT  PK   e.g. "RC-7F3K2"
  status             TEXT       draft | in_progress | complete | missed | closed
  fraud_at           DATETIME   UTC, earliest transaction time
  amount_total       INTEGER    rupees, sum of transactions
  transactions       JSON       [{amount, utr, at, app, method, payee_id, confidence{}}]
  bank, payment_app, method, payee_id   TEXT
  scam_type          TEXT       fake_kyc | job_task | investment | digital_arrest |
                                fake_customer_care | sextortion | loan_app | other
  scam_confidence    REAL       0..1
  incident_summary   TEXT       one-paragraph summary (user text or AI-cleaned)
  prior_compensation BOOLEAN
  language           TEXT       en | ta
  ack_1930, ack_1930_at, called_pending
  ack_portal, ack_portal_at
  bank_ref, bank_reported_at
  completed_at       DATETIME
  followup_state     TEXT       none | day30_no_reply | resolved | ombudsman_drafted
  source             TEXT       cache | ai | manual
  is_demo_seed       BOOLEAN
  created_at, updated_at
```

Derived on read (never stored): `eligibility` and `steps`, so a change in the clock or a rule constant is reflected immediately.

7. API reference

Method	Path	Purpose
GET	`/api/health`	Liveness, demo mode flag, current (demo) time
POST	`/api/extract`	Multipart `image` (≤ 10 MB, PNG/JPEG/WebP), `description`, `language` → fields, confidences, scam type, source
POST	`/api/cases`	Create a case from confirmed fields → `CaseOut`
GET	`/api/cases/{id}`	Case with recomputed eligibility and step states
PATCH	`/api/cases/{id}`	Save `ack_1930`, `called_pending`, `ack_portal`, `bank_ref`, `followup_state`

Method	Path	Purpose
GET	`/api/cases/{id}/docs`	Generated 1930 script, NCRP text (+ char count), bank subject/body, follow-up drafts
GET	`/api/cases/{id}/pack.pdf`	Case pack PDF
GET	`/api/cases`	Dashboard list, newest first
GET	`/api/stats`	Totals, compensation exposure, scam-type counts, due-in-24 h, missed, completion rate
POST	`/api/demo/clock`, `/api/demo/reset`	Demo-mode only: move the clock, reset and reseed

```
// CaseOut (abridged)
{
  "id": "RC-7F3K2", "status": "in_progress", "amount_total": 18000,
  "fraud_at": "2027-01-12T14:12:00Z", "scam_type": "fake_kyc", "scam_confidence": 0.92,
  "eligibility": { "status": "ELIGIBLE_PENDING", "max_compensation": 15300,
                   "deadline_at": "2027-01-17T14:12:00Z", "seconds_left": 430860,
                   "bank_resolve_by": "2027-02-26", "message_key": "elig.pending" },
  "steps": { "1930":   { "done": true,  "ref": "31012270045871", "at": "..." },
             "portal": { "done": false, "ref": null, "at": null },
             "bank":   { "done": false, "ref": null, "at": null } }
}
```

8. Eligibility engine

A pure function with all rule constants in one module, so policy changes are a one-line edit with tests.

```
RULE_EFFECTIVE_FROM = 2027-01-01   COMP_MAX_LOSS = 50,000   COMP_RATE = 0.85
COMP_CAP = 25,000                   REPORT_WINDOW_DAYS = 5   BANK_RESOLVE_DAYS = 45

deadline_at = fraud_at + 5 days
status = first match of:
  fraud_at.date < RULE_EFFECTIVE_FROM                    → NOT_APPLICABLE_DATE
  prior_compensation                                      → NOT_APPLICABLE_PRIOR
  amount_total > COMP_MAX_LOSS                           → NOT_APPLICABLE_AMOUNT
  now > deadline_at and not (NCRP and bank before deadline) → WINDOW_MISSED
  NCRP (1930 or portal) and bank both reported before deadline → ELIGIBLE_COMPLETE
  otherwise                                               → ELIGIBLE_PENDING
max_compensation = min(round(0.85 × amount_total), 25,000) for ELIGIBLE_*; else 0
```

Test case	Expected
₹18,000 on 12 Jan 2027, nothing reported, 19 min later	ELIGIBLE_PENDING · ₹15,300
Same, all three reported on day 1	ELIGIBLE_COMPLETE · ₹15,300
₹40,000	₹25,000 (cap)
₹60,000	NOT_APPLICABLE_AMOUNT
Fraud on 20 Dec 2026	NOT_APPLICABLE_DATE
Day 6, bank not reported	WINDOW_MISSED

> The engine produces an **estimate**. Final eligibility, including any negligence assessment, is the bank's decision; the UI states this.

9. AI evidence pipeline

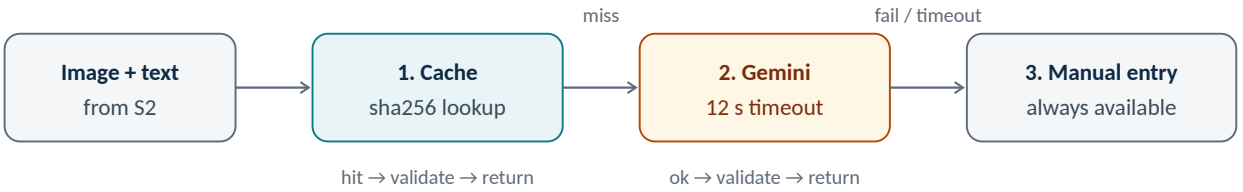


Figure 3. Fallback ladder — the demo can never stall on an AI call.

- **Single call:** the image and the user's description are sent together; the model returns extraction fields, per-field confidence, scam type, scam confidence and a one-sentence summary as schema-constrained JSON. Unknown values must be `null` — the prompt forbids guessing digits.
- **Validation:** UPI UTR must be 12 digits; amount must be a positive integer; UPI IDs must match `^[\w.\-]{2,256}@[a-zA-Z]{2,64}$`; failures lower confidence instead of rejecting.
- **Human in the loop:** fields under 0.7 confidence are highlighted, and nothing is generated until the user confirms every field.
- **Keyword fallback classifier:** if the AI is unavailable, keywords map the description to a scam type (for example *CBI*, *customs*, *arrest*, *parcel* → digital arrest) with confidence 0.6.
- **Model choice:** the model ID is configuration, not code, so it can be swapped for a paid or self-hosted model (e.g. Gemma) without changes.

10. Document generation

Official texts are produced from **deterministic templates**, not free-form AI output, so every number in them is exactly what the user confirmed. Only the incident summary may be AI-cleaned, and it falls back to the user's own words.

Document	Key rules
1930 call script	Amount, method, date-time, UTR, payee, bank, one-line summary; ends by asking for the acknowledgement number
NCRP description	At least 200 characters; characters `# $ @ ^ *` removed; includes the 1930 acknowledgement if available
Bank letter	Subject with amount and UTR; both acknowledgement numbers; requests hold, reference number and processing under the 2026 RBI directions; only last 4 account digits
Day-30 / Day-45	Reminder to bank; draft Ombudsman complaint marked "verify current process before submitting"

11. Security and privacy

11.1 Principles

- **Data minimisation:** collect only what the three reports need; never OTP, PIN, password, card number or full account number.
- **Purpose limitation and consent:** data is used only to prepare the victim's reports; a consent line is shown before upload.
- **User control:** delete-my-case removes the case and evidence.
- **No silent action:** Reclaim never submits to any government or bank system.

These follow the principles of India's Digital Personal Data Protection Act; formal compliance would be assessed before production.

11.2 Threat model

Threat	Mitigation
Victim tricked by a fake "Reclaim" asking for money	Distribution only via bank/police channels; persistent "we never charge or ask for OTPs" banner; recovery-scam warning
Sensitive data leakage to third-party AI	Prototype uses only fabricated sample data; production uses a paid enterprise tier or a self-hosted model inside the bank's network
Malicious uploads	MIME and size checks (≤ 10 MB, images only); files processed in memory, not executed or served back
Abuse of AI endpoint / cost	Rate limiting per IP; cache-first design
Data exposure via logs	Structured logs exclude images, descriptions and identifiers
Unauthorised dashboard access	Prototype is demo-only; production uses bank SSO with role-based access and audit logs
Secrets in repository	`.env` git-ignored; example files only

12. Reliability and demo safety

- **Demo clock:** the RBI rule applies from 1 Jan 2027, but the event is in Oct 2026. In demo mode, `now()` starts at 12 Jan 2027, 8:01 PM IST and advances in real time; the stage can jump to day 4, day 30 or reset.
- **Offline operation:** fixtures cover all sample screenshots; the full flow runs with Wi-Fi off and no API key.
- **Seeded dashboard:** about 12 fabricated cases across scam types and deadline states, so the bank view is never empty.
- **Polling over websockets:** a 2-second poll is simpler and survives reconnects.
- **Time handling:** stored in UTC, displayed in Asia/Kolkata, tested around midnight boundaries.

13. Production architecture (target)

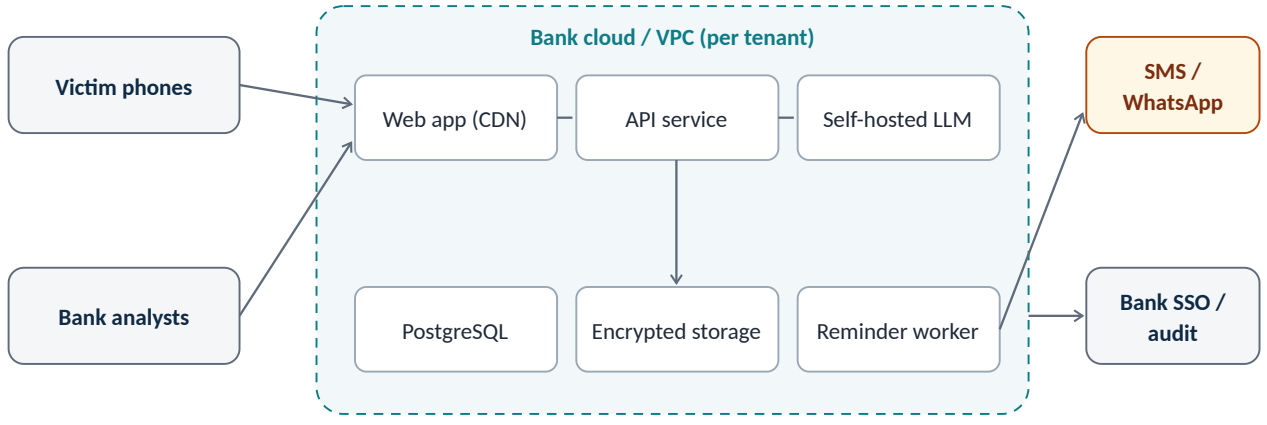


Figure 4. Target deployment — each bank runs Reclaim in its own environment so victim data stays with the bank.

Concern	Prototype	Production
Database	SQLite	PostgreSQL with encryption at rest, per-tenant isolation
Evidence files	In memory	Encrypted object storage with retention policy
AI	Gemini free tier, fake data	Paid enterprise tier or self-hosted model in the bank's network

Concern	Prototype	Production
Reminders	Simulated in UI	Background worker + SMS/WhatsApp provider
Dashboard auth	None (demo)	Bank SSO, role-based access, audit trail
Integration	Standalone link	Deep link from the bank app's "Report fraud"; case export API into the bank's complaint system

14. Testing strategy

Level	What
Unit	Eligibility engine (all rule paths), field validation and masking, document templates (≥ 200 chars, forbidden characters removed), keyword classifier
Fixture	Each sample screenshot's extraction compared field-by-field with its expected JSON
API	Create → read → patch → complete; eligibility recomputed after demo-clock jumps
End-to-end	Scripted run of the demo story on a 360 px viewport; dashboard reflects each step within 2 s
Failure	No network, no API key, unknown image, oversized file, invalid acknowledgement numbers

15. Limitations and roadmap

- No public API exists to submit NCRP complaints; filing remains a user action by design.
- Eligibility logic is based on secondary summaries of the RBI directions and must be checked against the official notification; rules for non-commercial banks may differ.
- Extraction accuracy depends on screenshot quality; the confirm step is mandatory for that reason.
- **Roadmap:** more Indian languages, WhatsApp-native flow, bank complaint-system integrations, insurer claim-pack API, aggregated fraud-trend reports for banks and law enforcement.

16. Glossary and references

Term	Meaning
1930	National cyber-fraud helpline; issues an acknowledgement number for financial-fraud reports
NCRP	National Cyber Crime Reporting Portal, cybercrime.gov.in
UTR	Unique Transaction Reference; 12 digits for UPI payments
Case pack	Reclaim's PDF bundle of a case's details, references and texts

References: RBI (Commercial Banks – Responsible Business Conduct) Third Amendment Directions, 2026 (summaries by ELP Law and corplawupdates.in) · LocalCircles UPI-fraud survey via Business Standard, Jun 2025 · the420.in, Odisha cyber-fraud data 2023–25 and 1930 helpline explainer · RBL Bank guide to NCRP reporting · Gemini API free-tier terms summary, agentdeals.dev.